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P.A. NAVY-AS

Other Cust ARMY-EL

International interest

Procurement Specification MIL-C-81703

DD FORM 672 10 APR 64 (Continuation) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

TITLE

CONNECTOR, PLUG, ELECTRIC, SOLDER TYPE, PUSH-PULL COUPLING (CLASS E, P AND J)

SERIES I

SUPERSEDES:

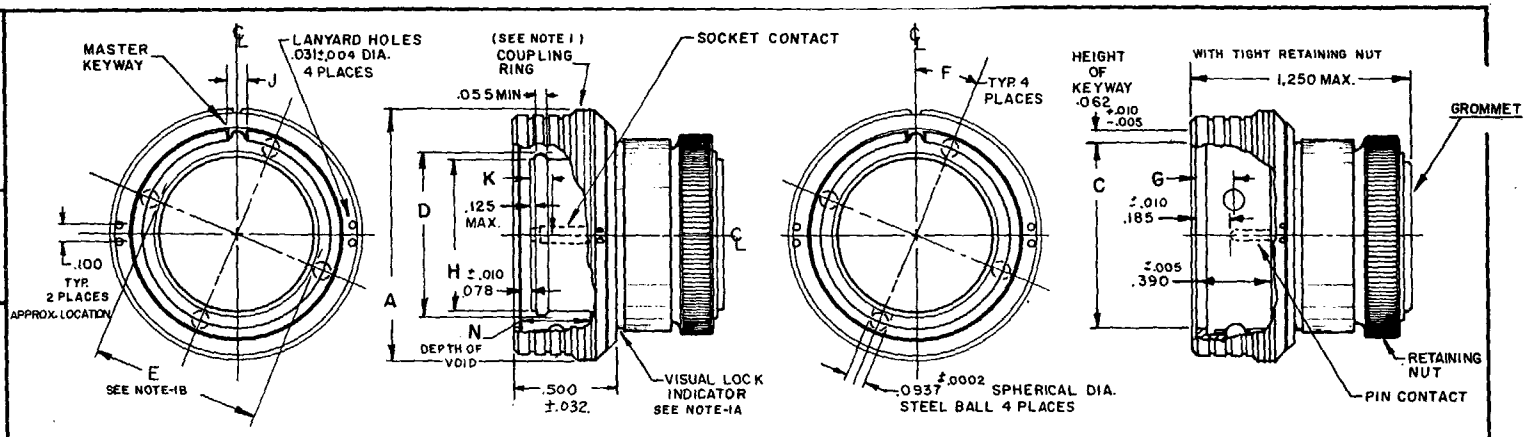
MILITARY STANDARD

MS 3137

SHEET 1 OF 2

5935-1761-4

FED. SUP CLASS 5935




STYLE S (SOCKET INSERT) CLASS E STYLE P (PIN INSERT)

| SHELL SIZE DASH NO. | A MAX. DIA. COUPLING RING | C MIN I.D. SHELL | D ±.006 AVERAGE DIA. INSERT SEAL | E MIN. DIA. BALL SPACING | F BALL LOCATION (ANGULAR) | G +.002 -.005 BALL LOCATION (DEPTH) | H ±.010 FRONT O.D. OF INSERT | J MAX WIDTH OF KEYWAY | N ±.031 DEPTH OF VOID | K POINT OF SPRING CONTACT | FOR INSERT ARRANGEMENT SEE DWG. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| -3 | .796 | .444 | .374 | .391 | 30° | .202 | .344 | .114 | IN DISENGAGED POSITION .310 | .138 MAX. MEASURE WITH SQUARE NOSE PIN HAVING BASIC DIAMETER OF MATING CONTACT | MS 33698 |
| -7 | .933 | .579 | .490 | .526 | 22½° | | .460 | | | | MS 33699 |
| -12 | 1.140 | .713 | .630 | .661 | | | .600 | | | | MS 33700 |
| -19 | 1.228 | .852 | .770 | .799 | | | .740 | | | | MS 33701 |
| -27 | 1.383 | 1.007 | .925 | .954 | | | .895 | | | | MS 33702 |
| -37 | 1.513 | 1.127 | 1.047 | 1.076 | | | 1.017 | | | | MS 33703 |
| | | | | | | | | | | | |

DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.016, ANGULAR TOLERANCE ±1/2°

NOTES

1. COUPLING RING AND BALLS ARE SHOWN IN THE ENGAGED POSITION FOR CLARITY. THIS POSITION SHALL ONLY BE POSSIBLE WHEN ENGAGED TO THE RECEPTACLE. IN THE NORMAL DISENGAGED POSITION THE FOLLOWING IS TRUE:
   A. THE VISUAL LOCK INDICATOR IS NOT IN VIEW.
   B. THE BALLS ARE RELEASED WHEN "E" MIN. EQUALS "C".

NOTE 2

EXAMPLE OF PART NO. MS3137 E 19-50 S Y-

MS NUMBER | CLASS | SIZE | INSERT ARRANGEMENT | CONTACT STYLE | ALTERNATE INSERT POSITION

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

APPROVED 14 MAY 59 REVISED (A) 10 SEPT. 62 (B) 1 NOV 1965 (C) 6 Mar 67 (D) 20 JAN 1969 (E) 21 JULY 1971

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DD FORM 10 APR 64 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

P.A. NAVY-AS
Other Cust ARMY-EL
Procurement Specification MIL-C-81703
International Interest

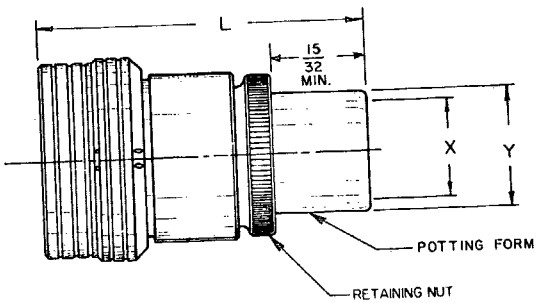
SUPERSEDES:
SERIES I (CLASS E P AND J)

TITLE
CONNECTOR, PLUG, ELECTRIC, SOLDER TYPE, PUSH-PULL COUPLING

MILITARY STANDARD
MS3137
SHEET 2
FED. SUP CLASS 5935

## CLASS P — POTTED SEAL

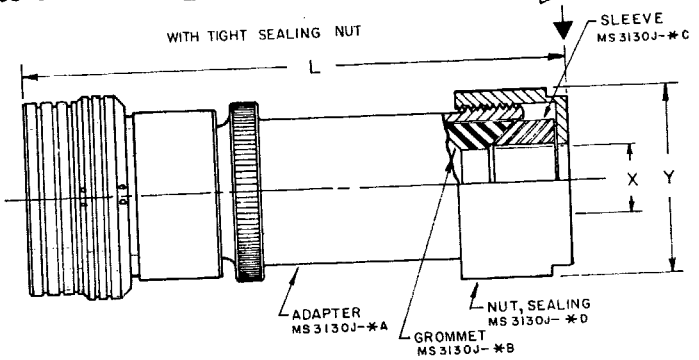
WITH TIGHT RETAINING NUT




| SHELL SIZE DASH NO. | L LENGTH MAX | X DIA. MIN I.D. POTTING FORM | Y DIA. MAX. O.D. POTTING FORM |
|---|---|---|---|
| -3 | 1.718 | .261 | .391 |
| -7 | | .372 | .515 |
| -12 | | .482 | .630 |
| -19 | | .602 | .760 |
| -27 | | .725 | .887 |
| -37 | | .857 | 1.047 |

## CLASS J — JACKETED CABLE GLAND SEAL (MP)

WITH TIGHT SEALING NUT




| SHELL SIZE DASH NO. | L LENGTH MAX. | X CABLE SEALING MIN. | X CABLE SEALING MAX. | Y MAX. DIA. NUT | ADAPTER MS3130J-*A | GROMMET MS3130J-*B | SLEEVE MS3130J-*C | NUT SEALING MS3130J-*D |
|---|---|---|---|---|---|---|---|---|
| -3 | 1.734 | .178 | .210 | .645 | MS3130J-3A | MS3130J-3B | MS3130J-3C | MS3130J-3D |
| -7 | 2.719 | .381 | .413 | .926 | MS3130J-7A | MS3130J-7B | MS3130J-7C | MS3130J-7D |
| -12 | 3.188 | .412 | .444 | .988 | MS3130J-12A | MS3130J-12B | MS3130J-12C | MS3130J-12D |
| -19 | 3.219 | .522 | .554 | 1.113 | MS3130J-19A | MS3130J-19B | MS3130J-19C | MS3130J-19D |
| -27 | 3.219 | .600 | .632 | 1.113 | MS3130J-27A | MS3130J-27B | MS3130J-27C | MS3130J-27D |
| -37 | 3.250 | .662 | .694 | 1.176 | MS3130J-37A | MS3130J-37B | MS3130J-37C | MS3130J-37D |

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.016, ANGULAR TOLERANCE ±1/2°

NOTES:
1. FOR CLASS J ADAPTER, SEALING NUT, SLEEVE AND GROMMET SEE MS3130 SHEET 3.

MP MAX. CABLE WEIGHTS IN POUNDS (ALLOWABLE FORCE IN DIRECTION INDICATED).

| SHELL SIZE | MAX. WEIGHTS |
|---|---|
| -3 | 2.0 |
| -7 | 2.7 |
| -12 | 4.0 |
| -19 | 5.5 |
| -27 | 8.0 |
| -37 | 12.0 |

EXAMPLE OF PART NO.

MS 3137 J 19-50 S Y

MS NUMBER | CLASS | SIZE | INSERT ARRANGEMENT | CONTACT STYLE | ALTERNATE INSERT POSITION

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APPROVED 14 MAY 59 REVISED (E) FOR CHANGES SEE SHEETS 1 & 2.

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